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Control device, control method, and computer-readable recording medium

Abstract

A control device for a moving body includes circuitry configured to: acquire an external environment recognition image representing a recognition result of a periphery of the moving body; and cause a display device to display an obstacle icon indicating an obstacle position based on an input from a user of the moving body in a superimposed manner on the external environment recognition image. The circuitry is further configured to: cause the display device to display a first external environment recognition image and a second external environment recognition image having different viewpoints as the external environment recognition image; and superimpose a first obstacle icon on the first external environment recognition image and superimpose a second obstacle icon different from the first obstacle icon on the second external environment recognition image, as the obstacle icon.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2022-039042 filed on Mar. 14, 2022, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a control device, a control method, and a computer-readable recording medium.

BACKGROUND ART

(3) In recent years, efforts have been actively made to provide access to sustainable transportation systems that are considerate of vulnerable people in transport participants. In order to realize the above object, attention is focused on research and development to further improve safety and convenience of traffic through research and development related to a driving assistance technology.

(4) The related art discloses a known technique for displaying a state of an external environment of a vehicle on a display device as a multi-view image when the vehicle is automatically parked in a parking space. In addition, a technique is known for displaying an obstacle present in a periphery of a vehicle as a virtual icon in an external environment image of the vehicle when the external environment image of the vehicle is displayed on a display device. International Patent Publication No. WO2017/028849 pamphlet (hereinafter, referred to as Patent Literature 1) discloses that an obstacle can be added to and deleted from, by a manual input such as a tap operation or a slide operation of a user, an automatic parking trajectory determined in advance, and the automatic parking trajectory adjusted by a manual operation is determined to park a vehicle.

(5) According to Patent Literature 1, when a user recognizes an obstacle that cannot be detected by a sensor or the like, it is possible to calculate an automatic parking trajectory in consideration of the obstacle that can be added by a manual operation input.

(6) However, Patent Literature 1 does not describe display of an obstacle in a case where an image of an external environment of the vehicle is displayed as a multi-view image. Therefore, in the driving assistance technology, there is room for improvement in the related-art technique regarding a method of displaying an obstacle displayed in an external environment image of a vehicle.

(7) The present disclosure provides a control device, a control method, and a computer-readable

recording medium which enable to display an obstacle in a periphery of a vehicle according to a viewpoint of an external environment recognition image of the vehicle. Further, the present disclosure contributes to development of sustainable transportation systems.

SUMMARY

(8) A first aspect of the present disclosure relates to a control device for a moving body, the control device including circuitry configured to: acquire an external environment recognition image representing a recognition result of a periphery of the moving body; and cause a display device to display an obstacle icon indicating an obstacle position based on an input from a user of the moving body in a superimposed manner on the external environment recognition image, in which the circuitry is configured to: cause the display device to display a first external environment recognition image and a second external environment recognition image having different viewpoints as the external environment recognition image; and superimpose a first obstacle icon on the first external environment recognition image and superimpose a second obstacle icon different from the first obstacle icon on the second external environment recognition image, as the obstacle icon.

(9) A second aspect of the present disclosure relates to a control method executed by a control device for a moving body, the control method including: acquiring an external environment recognition image representing a recognition result of a periphery of the moving body; causing a display device to display an obstacle icon indicating an obstacle position based on an input from a user of the moving body in a superimposed manner on the external environment recognition image; causing the display device to display a first external environment recognition image and a second external environment recognition image having different viewpoints as the external environment recognition image; and superimposing a first obstacle icon on the first external environment recognition image and superimposing a second obstacle icon different from the first obstacle icon on the second external environment recognition image as the obstacle icon.

(10) A third aspect of the present disclosure relates to a non-transitory computer-readable recording medium storing a control program for causing a processor of a control device for a moving body to execute processing, the processing including: acquiring an external environment recognition image representing a recognition result of a periphery of the moving body, causing a display device to display an obstacle icon indicating an obstacle position based on an input from a user of the moving body in a superimposed manner on the external environment recognition image, causing the display device to display a first external environment recognition image and a second external environment recognition image having different viewpoints as the external environment recognition image, and superimposing a first obstacle icon on the first external environment recognition image and superimposing a second obstacle icon different from the first obstacle icon on the second external environment recognition image as the obstacle icon.

(11) According to the present disclosure, it is possible to provide a control device, a control method, and a computer-readable recording medium which enable to display an obstacle in a periphery of a vehicle according to a viewpoint of an external environment recognition image of the vehicle.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Exemplary embodiment(s) of the present invention will be described in detail based on the following figures, wherein:

(2) FIG. 1 is a side view showing an example of a vehicle on which a control device according to an embodiment is mounted:

(3) FIG. 2 is a top view of the vehicle shown in FIG. 1;

- (4) FIG. 3 is a block diagram showing an internal configuration of the vehicle shown in FIG. 1;
- (5) FIG. 4 is a flow chart showing an example of display control for an external environment recognition image;
- (6) FIG. 5 is a view showing an example of a top view obstacle icon displayed in a top view image;
- (7) FIG. 6 is a view showing an example of a three-dimensional obstacle icon displayed in a three-dimensional image;
- (8) FIG. 7 is a flow chart showing a modification of the display control of the external environment recognition image;
- (9) FIG. 8 is a view showing an example of a combined image including the top view image and the three-dimensional image; and
- (10) FIG. 9 is a view showing a modification of top view obstacle icons displayed in the top view image.

DESCRIPTION OF EMBODIMENTS

(11) Hereinafter, an embodiment of a control device, a control method, and a computer-readable recording medium according to the present disclosure will be described with reference to the accompanying drawings. The drawings are viewed in directions of reference numerals. In addition, in the present specification and the like, in order to simplify and clarify the description, a front-rear direction, a left-right direction, and an upper-lower direction are described according to directions viewed from a driver of a vehicle **10** shown in FIGS. 1 and 2. In the drawings, a front side of the vehicle **10** is denoted by Fr, a rear side thereof is denoted by Rr, a left side thereof is denoted by L, a right side thereof is denoted by R, an upper side thereof is denoted by U, and a lower side thereof is denoted by D.

(12) <Vehicle **10** on which Control Device of Present Disclosure is Mounted>

(13) FIG. 1 is a side view of the vehicle **10** on which a control device according to the present disclosure is mounted. FIG. 2 is a top view of the vehicle **10** shown in FIG. 1. The vehicle **10** is an example of a moving body in the present disclosure.

(14) The vehicle **10** is an automobile including a drive source (not shown) and wheels including driving wheels driven by power of the drive source and steering wheels that are steerable. In the present embodiment, the vehicle **10** is a four-wheeled automobile including a pair of left and right front wheels and a pair of left and right rear wheels. The drive source of the vehicle **10** is, for example, an electric motor. The drive source of the vehicle **10** may be an internal combustion engine such as a gasoline engine or a diesel engine, or a combination of an electric motor and an internal combustion engine. In addition, the drive source of the vehicle **10** may drive the pair of left and right front wheels, the pair of left and right rear wheels, or four wheels, that is, the pair of left and right front wheels and the pair of left and right rear wheels. The front wheels and the rear wheels may both be steering wheels that are steerable, or the front wheels or the rear wheels may be steering wheels that are steerable.

(15) The vehicle **10** further includes side mirrors **11L** and **11R**. The side mirrors **11L** and **11R** are mirrors (rearview mirrors) that are provided at outer sides of front seat doors of the vehicle **10** and that allow the driver to check a rear side and rear lateral sides. Each of the side mirrors **11L** and **11R** is fixed to a body of the vehicle **10** by a rotation shaft extending in a vertical direction and can be opened and closed by rotating about the rotation shaft.

(16) The vehicle **10** further includes a front camera **12Fr**, a rear camera **12Rr**, a left side camera **12L**, and a right side camera **12R**. The front camera **12Fr** is a digital camera that is provided at a front portion of the vehicle **10** and captures an image of a front side with respect to the vehicle **10**. The rear camera **12Rr** is a digital camera that is provided at a rear portion of the vehicle **10** and captures an image of a rear side with respect to the vehicle **10**. The left side camera **12L** is a digital camera that is provided on the left side mirror **11L** of the vehicle **10** and captures an image of a left side with respect to the vehicle **10**. The right side camera **12R** is a digital camera that is provided on the right side mirror **11R** of the vehicle **10** and captures an image of a right side with respect to

the vehicle **10**.

(17) <Internal Configuration of Vehicle **10**>

(18) FIG. **3** is a block diagram showing an example of an internal configuration of the vehicle **10** shown in FIG. **1**. As shown in FIG. **3**, the vehicle **10** includes a sensor group **16**, a navigation device **18**, a control electronic control unit (ECU) **20**, an electric power steering (EPS) system **22**, and a communication unit **24**. The vehicle **10** further includes a driving force control system **26** and a braking force control system **28**.

(19) The sensor group **16** acquires various detection values used for control performed by the control ECU **20**. The sensor group **16** includes the front camera **12Fr**, the rear camera **12Rr**, the left side camera **12L**, and the right side camera **12R**. The sensor group **16** includes a front sonar group **32a**, a rear sonar group **32b**, a left side sonar group **32c**, and a right side sonar group **32d**. The sensor group **16** includes wheel sensors **34a** and **34b**, a vehicle speed sensor **36**, and an operation detection unit **38**. The sensor group **16** may include a radar.

(20) The front camera **12Fr**, the rear camera **12Rr**, the left side camera **12L**, and the right side camera **12R** acquire recognition data (for example, an external environment recognition image) for recognizing a periphery of the vehicle **10** by capturing images of the periphery of the vehicle **10**. Peripheral images captured by the front camera **12Fr**, the rear camera **2Rr**, the left side camera **12L**, and the right side camera **12R** are referred to as a front image, a rear image, a left side image, and a right side image, respectively. An image formed by the left side image and the right side image may be referred to as a side image. The external environment recognition image generated by synthesizing respective pieces of imaging data of the front camera **12Fr**, the rear camera **12Rr**, the left side camera **12L**, and the right side camera **12R** may be referred to as a top view image of the vehicle **10**.

(21) The front sonar group **32a**, the rear sonar group **32b**, the left side sonar group **32c**, and the right side sonar group **32d** emit sound waves to the periphery of the vehicle **10** and receive reflected sounds from other objects. The front sonar group **32a** includes, for example, four sonars. The sonars constituting the front sonar group **32a** are provided on an obliquely left front side, a front left side, a front right side, and an obliquely right front side of the vehicle **10**, respectively. The rear sonar group **32b** includes, for example, four sonars. The sonars constituting the rear sonar group **32b** are provided on an obliquely left rear side, a rear left side, a rear right side, and an obliquely right rear side of the vehicle **10**, respectively. The left side sonar group **32c** includes, for example, two sonars. The sonars constituting the left side sonar group **32c** are provided on a front side and a rear side of a left side portion of the vehicle **10**, respectively. The right side sonar group **32d** includes, for example, two sonars. The sonars constituting the right side sonar group **32d** are provided on a front side and a rear side of a right side portion of the vehicle **10**, respectively.

(22) The wheel sensors **34a** and **34b** detect a rotation angle of the wheel of the vehicle **10**. The wheel sensors **34a** and **34b** may be implemented by angle sensors or displacement sensors. The wheel sensors **34a** and **34b** output detection pulses each time the wheel rotates by a predetermined angle. The detection pulses output from the wheel sensors **34a** and **34b** are used to calculate the rotation angle of the wheel and a rotation speed of the wheel. A movement distance of the vehicle **10** is calculated based on the rotation angle of the wheel. The wheel sensor **34a** detects, for example, a rotation angle θ_a of the left rear wheel. The wheel sensor **34b** detects, for example, a rotation angle θ_b of the right rear wheel.

(23) The vehicle speed sensor **36** detects a speed of a vehicle body of the vehicle **10**, that is, a vehicle speed V . and outputs the detected vehicle speed V to the control ECU **20**. The vehicle speed sensor **36** detects the vehicle speed V based on, for example, rotation of a countershaft of a transmission.

(24) The operation detection unit **38** detects a content of an operation performed by a user using an operation input unit **14** and outputs the detected content of the operation to the control ECU **20**. The operation input unit **14** includes, for example, various user interfaces such as a side mirror

switch for switching between an opened state and a closed state of the side mirrors **11L** and **11R**, and a shift lever (a select lever or a selector).

(25) The navigation device **18** detects a current position of the vehicle **10** by using, for example, a global positioning system (GPS), and guides the user to a path to a destination. The navigation device **18** includes a storage device (not shown) that includes a map information database.

(26) The navigation device **18** includes a touch panel **42** and a speaker **44**. The touch panel **42** functions as an input device and a display device of the control ECU **20**. The speaker **44** outputs various types of guidance information to an occupant of the vehicle **10** by voice.

(27) The touch panel **42** is configured to input various commands to the control ECU **20**. For example, the user can input a command related to parking assistance of the vehicle **10** via the touch panel **42**. The touch panel **42** is configured to display various screens related to a control content of the control ECU **20**. For example, a screen related to the parking assistance of the vehicle **10** is displayed on the touch panel **42**. Specifically, the touch panel **42** displays a parking assistance button for requesting the parking assistance of the vehicle **10**. The parking assistance button includes an automatic parking button for requesting parking by automatic steering of the control ECU **20** and a parking guidance button for requesting guidance when parking by an operation of the driver. The touch panel **42** displays the external environment recognition image for recognizing the periphery of the vehicle **10**. Constituent elements other than the touch panel **42**, for example, a head-up display (HUD), a smartphone, or a tablet terminal may be used as the input device or the display device.

(28) The “parking” is synonymous with, for example, “parking the moving body”. For example, the “parking” is a stop accompanied by getting on and off of an occupant, and excludes a temporary stop due to a traffic signal or the like. The “parking position” is a position at which the moving body is stopped, that is, a position for parking.

(29) The control ECU **20** includes an input and output unit **50**, a calculation unit **52**, and a storage unit **54**. The calculation unit **52** is implemented by, for example, a central processing unit (CPU). The calculation unit **52** performs various types of control by controlling units based on a program stored in the storage unit **54**. Further, the calculation unit **52** inputs and outputs a signal from and to units connected to the control ECU **20** via the input and output unit **50**.

(30) The calculation unit **52** includes an external environment recognition unit **55** for recognizing the external environment recognition image, a parking control unit **56** for controlling automatic parking of the vehicle **10**, and a display control unit **57** for performing display control of the external environment recognition image. The calculation unit **52** is an example of the control device according to the present disclosure.

(31) The external environment recognition unit **55** acquires, from the cameras, the external environment recognition image representing a recognition result of the periphery of the vehicle **10** captured by the front camera **12Fr**, the rear camera **12Rr**, the left side camera **12L**, and the right side camera **12R**.

(32) The parking control unit **56** performs the parking assistance of the vehicle **10** by automatic steering in which a steering **110** is automatically operated under control of the parking control unit **56**. In the assistance of the automatic steering, an accelerator pedal (not shown), a brake pedal (not shown), and the operation input unit **14** are automatically operated. The parking control unit **56** performs guide assistance when the driver manually parks the vehicle **10** by operating the accelerator pedal, the brake pedal, and the operation input unit **14**.

(33) For example, based on recognition data of an external environment of the vehicle **10** acquired by the front camera **12Fr**, the rear camera **12Rr**, the left side camera **12L**, and the right side camera **12R**, a predetermined parking position (hereinafter, also referred to as a “target parking position”) designated by the user, and a position of an object (hereinafter, also referred to as an “obstacle position”) that is present in a periphery of the target parking position and that is designated as an obstacle by the user, the parking control unit **56** controls automatic parking for parking the vehicle

10 at the target parking position.

(34) The object designated as an obstacle by the user is an object that does not actually become an obstacle for the automatic parking, that is, an object that is not detected as an obstacle by the parking control unit **56** but that the user desires to include as an obstacle. The object that the user desires to include as an obstacle is, for example, an object that the vehicle **10** is not desired to pass at the time of the automatic parking. Specifically, the object that the vehicle **10** is not desired to pass is a “puddle”, “dust”, a “native-grown plant”, or the like that is present on a movement path of the automatic parking of the vehicle **10**. Further, the obstacle designated by the user may include, for example, an area that is set around a place where a child is present and that has a radius $R2$ larger than a range of a radius $R1$ actually determined not to be an obstacle for the automatic parking ($R1 < R2$).

(35) The parking control unit **56** receives the obstacle position designated by the user. The obstacle position is designated by, for example, an input operation of the user on the touch panel **42**. The parking control unit **56** registers the target parking position and the obstacle position designated with respect to the target parking position in the storage unit **54** in association with each other.

(36) The parking control unit **56** notifies the user of candidates for the obstacle position at the target parking position, and receives designation of the obstacle position selected by the user from the candidates for the obstacle position. The candidates for the obstacle position at the target parking position refer to candidates for a position of an object that may be expected to become an obstacle at the target parking position, that is, a position of an obstacle that is uniquely determined based on the target parking position. The parking control unit **56** displays and notifies the user of a candidate for the obstacle position selected by the user and a candidate for the obstacle position not selected by the user in the candidates for the obstacle position in different modes (for example, different colors or different transparency levels).

(37) The display control unit **57** generates the external environment recognition image of the vehicle **10** based on the imaging data acquired by the cameras of the vehicle **10**. Specifically, the display control unit **57** generates a synthesized image by synthesizing the respective pieces of imaging data acquired by the front camera **12Fr**, the rear camera **12Rr**, the left side camera **12L**, and the right side camera **12R**, and generates a top view image of the vehicle **10** and the periphery of the vehicle **10** showing a state in which the synthesized image is viewed from above. The top view image is an example of a first external environment recognition image in the present disclosure.

(38) The display control unit **57** performs image processing of three-dimensionally reconstructing the synthesized image of the respective pieces of imaging data acquired by the front camera **12Fr**, the rear camera **12Rr**, the left side camera **12L**, and the right side camera **12R**, and generates a three-dimensional image virtually showing a state in which the vehicle **10** and the periphery of the vehicle **10** are rotated and viewed from, for example, a side or an obliquely upper side. The three-dimensional image is an example of a second external environment recognition image in the present disclosure. The first external environment recognition image and the second external environment recognition image are external environment recognition images having different viewpoints.

(39) The display control unit **57** displays the generated external environment recognition image on the display device of the vehicle **10**. Specifically, the display control unit **57** causes the touch panel **42** to display the top view image and the three-dimensional image of the vehicle **10** generated by synthesizing the respective pieces of imaging data of the front camera **12Fr**, the rear camera **12Rr**, the left side camera **12L**, and the right side camera **12R**.

(40) The display control unit **57** displays an obstacle icon indicating an obstacle position based on an input from the user of the vehicle **10** on the touch panel **42** in a superimposed manner on the external environment recognition image (the top view image and the three-dimensional image). The display control unit **57** superimposes an image of an obstacle icon of a predetermined virtual

object viewed from a viewpoint of the top view image (hereinafter, also referred to as a top view obstacle icon) on the top view image, and superimposes an image of an obstacle icon of the predetermined virtual object viewed from a viewpoint of the three-dimensional image (hereinafter, also referred to as a three-dimensional obstacle icon) on the three-dimensional image for display. The top view obstacle icon and the three-dimensional obstacle icon are images indicating a position of the same obstacle in the periphery of the vehicle **10**. The top view obstacle icon is an image of the virtual object viewed from above, and the three-dimensional obstacle icon is an image of the virtual object viewed from the side, the obliquely upper side, or the like. The image of the top view obstacle icon and the image of the three-dimensional obstacle icon are stored in the storage unit **54** as obstacle images created in advance. The obstacle image may be, for example, an image imitating a pylon. The top view obstacle icon is an example of a first obstacle icon in the present disclosure. The three-dimensional obstacle icon is an example of a second obstacle icon in the present disclosure.

(41) The display control unit **57** switches, based on a predetermined input from the user, for example, an operation input from the touch panel **42** or an ON operation of a switch, display of the top view image and the three-dimensional image to be displayed on the touch panel **42**. The display switching between the top view image and the three-dimensional image may be performed based on, for example, a movement speed of the vehicle **10** or a shift position of a gear. In accordance with the external environment recognition image that changes with movement of the vehicle **10**, the display control unit **57** changes a superimposed position of the obstacle icon to be superimposed and displayed on the external environment recognition image.

(42) The EPS system **22** includes a steering angle sensor **100**, a torque sensor **102**, an EPS motor **104**, a resolver **106**, and an EPS ECU **108**. The steering angle sensor **100** detects a steering angle θ_{st} of the steering **110**. The torque sensor **102** detects a torque TQ applied to the steering **110**.

(43) The EPS motor **104** applies a driving force or a reaction force to a steering column **112** connected to the steering **110**, thereby enabling assistance of an operation performed by the occupant on the steering **110** and enabling the automatic steering during the parking assistance. The resolver **106** detects a rotation angle θ_m of the EPS motor **104**. The EPS ECU **108** controls the entire EPS system **22**. The EPS ECU **108** includes an input and output unit (not shown), a calculation unit (not shown), and a storage unit (not shown).

(44) The communication unit **24** enables wireless communication with another communication device **120**. The other communication device **120** is a base station, a communication device of another vehicle, an information terminal such as a smartphone carried by the occupant of the vehicle **10**, or the like.

(45) The driving force control system **26** includes a driving ECU **130**. The driving force control system **26** executes driving force control of the vehicle **10**. The driving ECU **130** controls a driving force of the vehicle **10** by controlling an engine (not shown) or the like based on an operation performed by the user on the accelerator pedal (not shown).

(46) The braking force control system **28** includes a braking ECU **132**. The braking force control system **28** executes braking force control of the vehicle **10**. The braking ECU **132** controls a braking force of the vehicle **10** by controlling a brake mechanism (not shown) or the like based on an operation performed by the user on the brake pedal (not shown).

(47) <Display Control of External Environment Recognition Image Performed by Calculation Unit **52**>

(48) Next, an example of display control of an external environment recognition image performed by the calculation unit **52** will be described based on a flow chart shown in FIG. **4**.

(49) For example, it is assumed that a user who drives the vehicle **10** arrives near a parking lot and taps the automatic parking button (not shown) on the touch panel **42** in order to automatically park the vehicle **10**. When the automatic parking button is tapped, external environment recognition images each representing the periphery of the vehicle **10** are displayed on the touch panel **42**, and at

that time, a selection screen is displayed on which it is possible to select whether to display the external environment recognition images as a top view image or as a three-dimensional image. When the user selects one of the external environment recognition images on the selection screen, the calculation unit 52 starts processing shown in FIG. 4.

(50) First, the display control unit 57 of the calculation unit 52 acquires imaging data of the periphery of the vehicle 10 captured by the front camera 12Fr, the rear camera 12Rr, the left side camera 12L, and the right side camera 12R (step S11).

(51) Next, the display control unit 57 determines whether to display the top view image as the external environment recognition image to be displayed on the touch panel 42, that is, whether the top view image is selected by the user on the selection screen (step S12).

(52) In step S12, when the top view image is selected (step S12: Yes), the display control unit 57 generates a top view image of the periphery of the vehicle 10 based on the imaging data acquired by the front camera 12Fr, the rear camera 12Rr, the left side camera 12L, and the right side camera 12R (step S13).

(53) Next, the display control unit 57 acquires the image of the top view obstacle icon, which is the obstacle icon for the top view image, from the storage unit 54 (step S14).

(54) Next, the display control unit 57 superimposes the top view obstacle icon acquired in step S14 on a predetermined position designated based on an operation input from the user in the top view image generated in step S13 (step S15). As described above, the top view obstacle icon is an image indicating an obstacle position designated by the user, and is displayed as an image of a virtual object viewed from above.

(55) Next, the display control unit 57 displays the top view image on which the top view obstacle icon is superimposed on the touch panel 42 (step S16). While the automatic parking is continued, the display control unit 57 returns to step S11 and repeats the processing. The display control unit 57 displays, on the touch panel 42, the top view image that changes in accordance with movement of the vehicle 10 in the automatic parking. The display control unit 57 changes a superimposed position of the top view obstacle icon in accordance with the change in the top view image caused by the movement of the vehicle 10.

(56) On the other hand, in step S12, when the top view image is not selected (step S12: No), the display control unit 57 determines whether to display the three-dimensional image as the external environment recognition image to be displayed on the touch panel 42, that is, whether the three-dimensional image is selected by the user on the selection screen (step S17).

(57) In step S17, when the three-dimensional image is not selected (step S17: No), the display control unit 57 returns to step S11 and repeats the processing.

(58) In step S17, when the three-dimensional image is selected (step S17: Yes), the display control unit 57 generates a three-dimensional image of the periphery of the vehicle 10 based on the imaging data acquired by the front camera 12Fr, the rear camera 12Rr, the left side camera 12L, and the right side camera 12R (step S18). The display control unit 57 may change a direction of the generated three-dimensional image based on, for example, an operation of the user. The display control unit 57 may generate the three-dimensional image as an image viewed from the front, the rear, the right, the left, or the like of the vehicle 10.

(59) Next, the display control unit 57 acquires the image of the three-dimensional obstacle icon, which is the obstacle icon for the three-dimensional image, from the storage unit 54 (step S19).

(60) Next, the display control unit 57 superimposes the three-dimensional obstacle icon acquired in step S19 on a predetermined position designated based on an operation input from the user in the three-dimensional image generated in step S18 (step S20). As described above, the three-dimensional obstacle icon is an image indicating the obstacle position designated by the user, and is displayed as an image of the virtual object viewed from, for example, a side.

(61) Next, the display control unit 57 displays the three-dimensional image on which the three-dimensional obstacle icon is superimposed on the touch panel 42 (step S21). While the automatic

parking is continued, the display control unit **57** returns to step **S11** and repeats the processing. The display control unit **57** displays, on the touch panel **42**, the three-dimensional image that changes in accordance with the movement of the vehicle **10** in the automatic parking. The display control unit **57** changes a superimposed position of the three-dimensional obstacle icon in accordance with the change in the three-dimensional image caused by the movement of the vehicle **10**.

(62) A mode of the external environment recognition image to be displayed on the touch panel **42** can be switched between the top view image and the three-dimensional image by the operation of the user. When the user performs a switching operation for the external environment recognition image, the display control unit **57** determines whether the external environment recognition image selected by the switching operation is the top view image or the three-dimensional image in step **S12** and step **S17**, and displays the determined external environment recognition image on the touch panel **42** according to the processing of each step.

(63) <Obstacle Icon Displayed in External Environment Recognition Image>

(64) Next, an obstacle icon displayed in an external environment recognition image (a top view image and a three-dimensional image) will be described with reference to FIGS. **5** and **6**.

(65) FIG. **5** is a view showing an example of a top view obstacle icon **68** displayed in a top view image **61**. As shown in FIG. **5**, the top view image **61** displayed on the touch panel **42** indicates an image of a situation in which a user automatically parks the vehicle **10** at a target parking position **66** of a home parking lot **62**. The home parking lot **62** is provided between a house (a building) **63** of the user and a plurality of plants (obstacles) **64a** to **64c**. The vehicle displayed in the top view image **61** is an image showing a state in which the vehicle **10** is viewed from above, and is a vehicle image **65** generated (captured) in advance and stored in the storage unit **54** or the like.

(66) In the top view image **61**, the top view obstacle icon **68** can be disposed, by a screen operation of the user, at an obstacle position designated by the user. For example, the top view obstacle icon **68** can be disposed at any position by performing a swipe operation on an obstacle marker **67** displayed at an upper corner of the top view image **61**. In the example shown in the drawing, the top view obstacle icon **68** is disposed in front of the building **63** close to the target parking position **66**. The top view obstacle icon **68** is displayed as an image imitating a pylon. The top view obstacle icon **68** is displayed as an image of the pylon viewed from above. A plurality of top view obstacle icons **68** may be arranged.

(67) In the top view image **61**, an operation panel **69** is displayed by which a position of the disposed top view obstacle icon **68** can be moved. A cursor is displayed on the operation panel **69**, and after the disposed top view obstacle icon **68** is touched and selected, the top view obstacle icon **68** can be moved in any arrow direction by the cursor of the operation panel **69**. On the operation panel **69**, a determination button **69a** for determining the disposition of the top view obstacle icon **68**, a reset button **69b** for canceling the disposition of the top view obstacle icon **68**, and a stop button **69c** for stopping the disposition of the top view obstacle icon **68** are displayed. The top view image **61** is an image corresponding to the top view image displayed in the processing of step **S16** in the display control shown in FIG. **4**.

(68) FIG. **6** is a view showing an example of a three-dimensional obstacle icon **72** displayed in a three-dimensional image **71**. As shown in FIG. **6**, the three-dimensional image **71** is a three-dimensional image of the periphery of the vehicle **10** shown in the top view image **61** in FIG. **5** viewed from a left rear side with respect to the vehicle **10**. The three-dimensional obstacle icon **72** in the three-dimensional image **71** is displayed as an image of the pylon as viewed from a side. Similarly to the top view obstacle icon **68** shown in FIG. **5**, the three-dimensional obstacle icon **72** is disposed in front of the building **63** close to the target parking position **66**. In setting of an obstacle position (a position of the three-dimensional obstacle icon **72**) in the three-dimensional image **71**, for example, when the obstacle position (the position of the top view obstacle icon **68**) in the top view image **61** is set by the designation of the user, the designated position in the top view image **61** is reflected. Also in the three-dimensional image **71**, the obstacle marker **67** and the

operation panel **69** may be displayed so that the disposition, movement, determination, and the like of the three-dimensional obstacle icon **72** can be performed. As in this example, a notification message **73** such as “DON'T DRIVE NEAR PYLON” or “PLEASE CHECK PERIPHERY DIRECTLY” may be displayed on the three-dimensional image **71**. The three-dimensional image **71** is an image corresponding to the three-dimensional image displayed in the processing of step **S21** in the display control shown in FIG. **4**.

(69) As described above, the display control unit **57** of the calculation unit **52** can cause the touch panel **42** to display, as the external environment recognition image, images having different viewpoints, for example, a top view image of the periphery of the vehicle **10** viewed from above and a three-dimensional image of the periphery of the vehicle **10** viewed from a side. The display control unit **57** superimposes the image of the top view obstacle icon in which the predetermined virtual object is viewed from the above on the top view image, and superimposes the image of the three-dimensional obstacle icon in which the predetermined virtual object is viewed from the side on the three-dimensional image, as obstacle icons indicating the same obstacle position designated based on the input from the user of the vehicle **10**. Accordingly, the top view obstacle icon corresponding to the viewpoint of the top view image is displayed in the top view image, and the three-dimensional obstacle icon corresponding to the viewpoint of the three-dimensional image is displayed in the three-dimensional image, and thus a sense of incongruity of the obstacle icon displayed in each image can be prevented.

(70) The display control unit **57** switches the display of the top view image and the three-dimensional image on the touch panel **42** based on, for example, an operation input to the touch panel **42** from the user. Accordingly, for each external environment recognition image selected by the user, an obstacle icon corresponding to a viewpoint of the external environment recognition image can be displayed, and the sense of incongruity of the obstacle icon displayed in each image can be prevented.

(71) <Modification of Display Control of External Environment Recognition Image>

(72) Next, a modification of the display control of the external environment recognition image performed by the calculation unit **52** will be described with reference to FIGS. **7** and **8**. FIG. **7** is a flow chart showing the modification of the display control of the external environment recognition image. FIG. **8** is a view showing an example of a combined image including the top view image and the three-dimensional image.

(73) As in the case of the processing in FIG. **4** described above, when the automatic parking button on the touch panel **42** is tapped and then an operation of selecting an external environment recognition image is performed, the calculation unit **52** starts processing shown in FIG. **7**.

(74) First, the display control unit **57** of the calculation unit **52** acquires imaging data of a periphery of the vehicle **10** captured by the front camera **12Fr**, the rear camera **12Rr**, the left side camera **12L**, and the right side camera **12R** (step **S31**).

(75) Next, the display control unit **57** generates a top view image of the periphery of the vehicle **10** based on the imaging data acquired in step **S31** (step **S32**). The top view image is, for example, the top view image **61** described with reference to FIG. **5**.

(76) Next, the display control unit **57** acquires a top view obstacle icon for a top view image in step **S33**, and superimposes the top view obstacle icon on the top view image in step **S34**. The processing contents are the same as those in steps **S14** and **S15** in FIG. **4**.

(77) Next, the display control unit **57** generates a three-dimensional image of the periphery of the vehicle **10** based on the imaging data acquired in step **S31** (step **S35**). The three-dimensional image is, for example, the three-dimensional image **71** described with reference to FIG. **6**.

(78) Next, the display control unit **57** acquires a three-dimensional obstacle icon for a three-dimensional image in step **S36**, and superimposes the three-dimensional obstacle icon on the three-dimensional image in step **S37**. The processing contents are the same as those in steps **S19** and **S20** in FIG. **4**.

(79) Next, the display control unit **57** generates a combined image in which the top view image generated in step **S34** on which the top view obstacle icon is superimposed and the three-dimensional image generated in step **S37** on which the three-dimensional obstacle icon is superimposed are displayed side by side (step **S38**).

(80) Next, the display control unit **57** displays the combined image generated in step **S38** on the touch panel **42** (step **S39**).

(81) For example, as shown in FIG. **8**, the external environment recognition image in the modification is displayed on the touch panel **42** as a combined image **81** in which the top view image **61** and the three-dimensional image **71** are arranged in a horizontal direction. The top view image **61** is the top view image described with reference to FIG. **5**, and superimposed with an image of a pylon viewed from above as the top view obstacle icon **68**. The three-dimensional image **71** is the three-dimensional image described with reference to FIG. **6**, and superimposed with a pylon viewed from a side as the three-dimensional obstacle icon **72**.

(82) While automatic parking is continued, the display control unit **57** returns to step **S31** and repeats the processing. The display control unit **57** displays, on the touch panel **42**, the combined image that includes the top view image and the three-dimensional image and that changes in accordance with movement of the vehicle **10** in the automatic parking. The display control unit **57** changes a superimposed position of the top view obstacle icon and a superimposed position of the three-dimensional obstacle icon which are superimposed on the respective images, in accordance with the change in the top view image and the three-dimensional image caused by the movement of the vehicle **10**.

(83) As described above, the display control unit **57** of the calculation unit **52** in the modification can cause the touch panel **42** to display side by side, as the external environment recognition image, images having different viewpoints, for example, a top view image of the periphery of the vehicle **10** viewed from above and a three-dimensional image of the periphery of the vehicle **10** viewed from a side. The display control unit **57** superimposes an image of a top view obstacle icon in which a predetermined virtual object is viewed from above on the top view image, and superimposes an image of a three-dimensional obstacle icon in which the predetermined virtual object is viewed from a side on the three-dimensional image, as obstacle icons indicating the same obstacle position designated based on an input from a user of the vehicle **10**. Accordingly, the obstacle icons (the top view obstacle icon and the three-dimensional obstacle icon) corresponding to viewpoints of the respective images are superimposed on the top view image and the three-dimensional image displayed side by side, and thus a sense of incongruity of the obstacle icon displayed in each image can be prevented.

(84) <Modification of Automatic Parking Control Performed by Calculation Unit **52**>

(85) Next, a modification of the automatic parking control performed by the parking control unit **56** of the calculation unit **52** will be described with reference to FIG. **9**. FIG. **9** is a view showing top view obstacle icons **88a** and **88b** displayed in the top view image **61**.

(86) In automatic parking control in the modification, as shown in FIG. **9**, the parking control unit **56** displays in advance, on the top view image **61**, the transparent top view obstacle icons **88a** and **88b** as candidates for an obstacle position, and performs a notification to the user. The candidates for the obstacle position refer to a position of an object that may be designated by the user as an obstacle, and refer to a position of an obstacle that is predicted and displayed by a parking control unit **56** side. Specifically, the parking control unit **56** predicts that the user may dispose a top view obstacle icon in front of the building **63** close to the target parking position **66**, and displays the transparent top view obstacle icon **88a** as a candidate. The parking control unit **56** predicts that the user may dispose the top view obstacle icon at a ground facing the target parking position **66**, and displays the transparent bar-shaped top view obstacle icon **88b** as a candidate.

(87) When the displayed top view obstacle icon **88a** or **88b** is tapped by the user, the parking control unit **56** receives a position of the tapped top view obstacle icon as an obstacle position

designated by the user. The parking control unit **56** changes a display state of the tapped top view obstacle icon from a transparent state to a non-transparent state for display. An operation method for moving the top view obstacle icon using the operation panel **69** displayed in the top view image **61** is the same as that described above. For example, the bar-shaped top view obstacle icon **88b** shown in FIG. **9** can be rotationally moved with a curved arrow cursor of the operation panel **69**. (88) As described above, the parking control unit **56** of the calculation unit **52** in the modification displays the candidates (the top view obstacle icons **88a** and **88b**) for the obstacle position on the touch panel **42** and performs the notification to the user, and receives the designation of the obstacle position selected by the user from the displayed candidates for the obstacle position. Accordingly, the obstacle position can be designated simply by tapping the top view obstacle icons **88a** and **88b** displayed as the candidates, a designation operation is made easy, and a burden at the time of setting the obstacle position can be reduced.

(89) Although the embodiment of the present disclosure has been described above, the present disclosure is not limited to the above embodiment, and modifications, improvements, and the like can be made as appropriate.

(90) For example, in the above embodiment, a case has been described where the top view image of the periphery of the vehicle **10** viewed from the above is displayed as the first external environment recognition image and the three-dimensional image of the periphery of the vehicle **10** viewed from the side is displayed as the second external environment recognition image, but the present disclosure is not limited thereto. For example, a three-dimensional image may be displayed as the first external environment recognition image, and a three-dimensional image having a viewpoint different from that of the three-dimensional image as the first external environment recognition image may be displayed as the second external environment recognition image. That is, three-dimensional images having different viewpoints may be switched and displayed on the touch panel **42**, or may be displayed side by side.

(91) In the above embodiment, the calculation unit **52** of the vehicle **10** has been described as the control device, but the present disclosure is not limited thereto. For example, the control device may be an information terminal such as a smartphone or a tablet terminal.

(92) In the above embodiment, a case has been described where the display control unit **57** displays the top view image and the three-dimensional image on the touch panel **42** of the vehicle **10**, but the present disclosure is not limited thereto. For example, the display control unit **57** may display the top view image and the three-dimensional image on a display screen of an information terminal (for example, a smartphone) carried by the occupant of the vehicle **10** via the communication unit **24**.

(93) The control method described in the above embodiment can be implemented by executing a control program prepared in advance on a computer. The control program is recorded in a computer-readable storage medium and is executed by being read from the storage medium. The control program may be provided in a form of being stored in a non-transitory storage medium such as a flash memory, or may be provided via a network such as the Internet. The computer that executes the control program may be provided in a control device, may be provided in an electronic device such as a smartphone, a tablet terminal, or a personal computer capable of communicating with the control device, or may be provided in a server device capable of communicating with the control device and the electronic device.

(94) At least the following matters are described in the present specification. Although corresponding constituent elements and the like in the above embodiment are shown in parentheses, the present disclosure is not limited thereto.

(95) (1) A control device (the calculation unit **52**) for a moving body (the vehicle **10**), the control device including: an external environment recognition unit (the external environment recognition unit **55**) configured to acquire an external environment recognition image representing a recognition result of a periphery of the moving body; and a display control unit (the display control

unit 57) configured to cause a display device (the touch panel 42) to display an obstacle icon indicating an obstacle position based on an input from a user of the moving body in a superimposed manner on the external environment recognition image, in which the display control unit is configured to cause the display device to display a first external environment recognition image (the top view image 61) and a second external environment recognition image (the three-dimensional image 71) having different viewpoints as the external environment recognition image, and superimpose a first obstacle icon (the top view obstacle icon 68) on the first external environment recognition image and superimpose a second obstacle icon (the three-dimensional obstacle icon 72) different from the first obstacle icon on the second external environment recognition image as the obstacle icon.

(96) According to (1), the obstacle icon corresponding to a viewpoint of the external environment recognition image is displayed, and thus a sense of incongruity of the obstacle icon can be prevented.

(97) (2) The control device according to (1), in which the first obstacle icon is an image of a predetermined virtual object viewed from a viewpoint of the first external environment recognition image, the second obstacle icon is an image of the predetermined virtual object viewed from a viewpoint of the second external environment recognition image, and the first obstacle icon and the second obstacle icon are images indicating the same position in the periphery of the moving body.

(98) According to (2), an image of the first obstacle icon and an image of the second obstacle icon when the predetermined virtual object is viewed from the viewpoints of the respective external environment recognition images are images indicating the same position, and thus the sense of incongruity of the obstacle icon can be prevented.

(99) (3) The control device according to (2), in which the first external environment recognition image is a top view image of the periphery of the moving body viewed from above, the first obstacle icon is an image of the predetermined virtual object viewed from above, the second external environment recognition image is a three-dimensional image of the periphery of the moving body viewed from a side, and the second obstacle icon is an image of the virtual object viewed from a side.

(100) According to (3), the obstacle icon in which the virtual object is viewed from the above is superimposed on the top view image of the periphery viewed from the above, and the obstacle icon in which the virtual object is viewed from the side is superimposed on the three-dimensional image of the periphery viewed from the side, and thus the sense of incongruity of the obstacle icon can be prevented.

(101) (4) The control device according to any one of (1) to (3), in which the display control unit is configured to switch display of the first external environment recognition image and the second external environment recognition image based on a predetermined input from the user.

(102) According to (4), for each external environment recognition image selected by the user, the obstacle icon corresponding to the viewpoint of the external environment recognition image can be displayed.

(103) (5) The control device according to any one of (1) to (4), further including: a parking control unit (the parking control unit 56) configured to perform parking control for parking the moving body at a target parking position based on the obstacle position.

(104) According to (5), the parking control can be executed in consideration of the obstacle position at which the vehicle is not desired to pass.

(105) (6) The control device according to (5), in which the parking control unit is configured to notify the user of candidates for the obstacle position, and receive designation of the obstacle position selected by the user from the candidates for the obstacle position.

(106) According to (6), the user can select any obstacle position from obstacle positions displayed as the candidates, and thus a designation operation for the obstacle position is made easy, and a burden at the time of setting the obstacle position can be reduced.

- (107) (7) The control device according to any one of (1) to (6), in which the display control unit is configured to change a superimposed position of the obstacle icon in response to a change in the external environment recognition image caused by movement of the moving body.
- (108) According to (7), the superimposed position of the obstacle icon changes in accordance with the change in the external environment recognition image caused by the movement of the vehicle, and thus the sense of incongruity of the obstacle icon can be prevented.
- (109) (8) A control method executed by a control device for a moving body, including: acquiring an external environment recognition image representing a recognition result of a periphery of the moving body; causing a display device to display an obstacle icon indicating an obstacle position based on an input from a user of the moving body in a superimposed manner on the external environment recognition image; causing the display device to display a first external environment recognition image and a second external environment recognition image having different viewpoints as the external environment recognition image; and superimposing a first obstacle icon on the first external environment recognition image and superimposing a second obstacle icon different from the first obstacle icon on the second external environment recognition image as the obstacle icon.
- (110) According to (8), the obstacle icon corresponding to a viewpoint of the external environment recognition image is displayed, and thus a sense of incongruity of the obstacle icon can be prevented.
- (111) (9) A non-transitory computer-readable recording medium storing a control program for causing a processor of a control device for a moving body to execute processing, the processing including: acquiring an external environment recognition image representing a recognition result of a periphery of the moving body; causing a display device to display an obstacle icon indicating an obstacle position based on an input from a user of the moving body in a superimposed manner on the external environment recognition image; causing the display device to display a first external environment recognition image and a second external environment recognition image having different viewpoints as the external environment recognition image; and superimposing a first obstacle icon on the first external environment recognition image and superimposing a second obstacle icon different from the first obstacle icon on the second external environment recognition image as the obstacle icon.
- (112) According to (9), the obstacle icon corresponding to a viewpoint of the external environment recognition image is displayed, and thus a sense of incongruity of the obstacle icon can be prevented.

Claims

1. A control device for a moving body, the control device comprising circuitry configured to: acquire an external environment recognition image representing a recognition result of a periphery of the moving body; and cause a display device to display an obstacle icon indicating an obstacle position based on an input from a user of the moving body in a superimposed manner on the external environment recognition image, wherein the circuitry is configured to: cause the display device to display a first external environment recognition image and a second external environment recognition image having different viewpoints as the external environment recognition image; and superimpose a first obstacle icon on the first external environment recognition image and superimpose a second obstacle icon different from the first obstacle icon on the second external environment recognition image, as the obstacle icon, and one of the first external environment recognition image or the second external environment recognition image is a virtually created view of an exterior of the moving body viewed from a side of the moving body.
2. The control device according to claim 1, wherein the first obstacle icon is an image of a predetermined virtual object viewed from a viewpoint of the first external environment recognition

image, the second obstacle icon is an image of the predetermined virtual object viewed from a viewpoint of the second external environment recognition image, and the first obstacle icon and the second obstacle icon are images indicating a same position in the periphery of the moving body.

3. The control device according to claim 2, wherein the first external environment recognition image is a top view image of the periphery of the moving body viewed from above, the first obstacle icon is an image of the predetermined virtual object viewed from above, the second external environment recognition image is a three-dimensional image of the periphery of the moving body viewed from a side, and the second obstacle icon is an image of the predetermined virtual object viewed from a side.

4. The control device according to claim 1, wherein the circuitry is configured to switch display of the first external environment recognition image and the second external environment recognition image based on a predetermined input from the user.

5. The control device according to claim 1, wherein the circuitry is further configured to: perform parking control for parking the moving body at a target parking position based on the obstacle position.

6. The control device according to claim 5, wherein the circuitry is configured to notify the user of candidates for the obstacle position, and receive designation of the obstacle position selected by the user from the candidates for the obstacle position.

7. The control device according to claim 1, wherein the circuitry is configured to change a superimposed position of the obstacle icon in response to a change in the external environment recognition image caused by movement of the moving body.

8. A control method executed by a control device for a moving body, the control method comprising: acquiring an external environment recognition image representing a recognition result of a periphery of the moving body; causing a display device to display an obstacle icon indicating an obstacle position based on an input from a user of the moving body in a superimposed manner on the external environment recognition image; causing the display device to display a first external environment recognition image and a second external environment recognition image having different viewpoints as the external environment recognition image; and superimposing a first obstacle icon on the first external environment recognition image and superimposing a second obstacle icon different from the first obstacle icon on the second external environment recognition image as the obstacle icon, wherein one of the first external environment recognition image or the second external environment recognition image is a virtually created view of an exterior of the moving body viewed from a side of the moving body.

9. A non-transitory computer-readable recording medium storing a control program for causing a processor of a control device for a moving body to execute processing, the processing comprising: acquiring an external environment recognition image representing a recognition result of a periphery of the moving body; causing a display device to display an obstacle icon indicating an] obstacle position based on an input from a user of the moving body in a superimposed manner on the external environment recognition image; causing the display device to display a first external environment recognition image and a second external environment recognition image having different viewpoints as the external environment recognition image; and superimposing a first obstacle icon on the first external environment recognition image and superimposing a second obstacle icon different from the first obstacle icon on the second external environment recognition image as the obstacle icon, wherein one of the first external environment recognition image or the second external environment recognition image is a virtually created view of an exterior of the moving body viewed from a side of the moving body.
